Multi-Year Localized Air Pollution Analysis and Forecasting

Point & Probabilistic Forecasting using Statistical and Deep Learning Models

Course: Applied Forecasting Methods (AFM)

Instructor: Pritam Anand

Institution: DAU Gandhinagar

Name	Roll Number
Harsh Jethwani	202518055
Anish Karmakar	202518061
Vrushil Panwala	202518010
Shah Trush	202518027

Submission Date: May 5, 2026

Abstract

Air pollution, particularly fine particulate matter ($PM_{2.5}$), poses a severe and ongoing public health threat, contributing to an estimated 7 million premature deaths annually worldwide. Accurate day-ahead $PM_{2.5}$ forecasting enables timely public health advisories, yet existing approaches typically provide only point estimates without any measure of uncertainty, a critical limitation for risk-aware decision making.

This report presents a comprehensive multi-year localized $PM_{2.5}$ forecasting study using daily data from 2001 to 2021 (7,670 observations), covering the full spectrum from classical statistical models to deep learning architectures and probabilistic quantile models. We evaluate seven classical time-series models (AR, MA, ARMA, ARIMA, SARIMA, ARIMAX, SARIMAX), three recurrent deep learning models (RNN, LSTM, GRU), and their quantile variants trained with pinball loss.

Keywords: $PM_{2.5}$ forecasting, time-series analysis, ARIMA, LSTM, GRU, quantile regression, pinball loss, prediction intervals, PICP, MPIW.

Contents

Abstract **1**

1 Introduction **4**

1.1 Background and Motivation 4

1.2 Problem Statement 5

1.3 Objectives 5

2 Data Source and Characteristics **6**

2.1 Data Origin 6

2.2 Data Characteristics 6

2.3 Data Pre-processing 7

2.3.1 Outlier Treatment 7

2.3.2 Calendar Continuity 7

2.3.3 Log-Transformation 7

2.3.4 Stationarity Testing and Differencing 9

2.3.5 Inverse-Transform Chain 9

2.3.6 Decomposition as a Diagnostic Tool 9

2.3.7 Feature Engineering (Deep Learning Models) 10

3 Exploratory Data Analysis **11**

3.1 Full Time Series 11

3.2 Distribution and Normality 12

3.3 Seasonal Structure 12

3.4 Feature Correlations 13

3.5 ACF and PACF of the Raw Series 13

4 Methodology **14**

4.1 Forecasting Setup 14

4.2 Model Order Selection 14

4.2.1 ACF/PACF Diagnostics 14

4.2.2 AIC Grid Search 15

4.2.3 Seasonal Order 16

4.3 Statistical Models 16

4.3.1 AR(p) — AutoRegressive Model 16

4.3.2 MA(q) — Moving Average Model . . . 16
4.3.3 ARMA(p, q) — Combined AR-MA . . . 16
4.3.4 ARIMA(p, d, q) — Integrated ARMA . . . 17
4.3.5 SARIMA$(p, d, q)(P, D, Q)_s$ — Seasonal ARIMA . . . 17
4.3.6 ARIMAX — ARIMA with Exogenous Variables . . . 17
4.3.7 SARIMAX — Full Seasonal ARIMA with Exogenous . . . 17
4.4 Evaluation Metrics . . . 17
4.5 Deep Learning Models . . . 18
4.5.1 Hyperparameter Tuning . . . 18
4.5.2 RNN — Recurrent Neural Network . . . 18
4.5.3 LSTM — Long Short-Term Memory . . . 19
4.5.4 GRU — Gated Recurrent Unit . . . 19
4.6 Quantile Models and Pinball Loss . . . 19
4.6.1 Evaluation Metrics: PICP and MPIW . . . 20

5 Analysis of Numerical Results 21
5.1 Baseline Models . . . 21
5.2 Statistical Models . . . 21
5.2.1 AR and MA Models . . . 21
5.2.2 ARMA and ARIMA Models . . . 22
5.2.3 Advanced Statistical Models . . . 22
5.3 Deep Learning Models . . . 24
5.4 Probabilistic (Quantile) Models . . . 25
5.5 Research Gap Identified . . . 26

6 Converting Forecasts into Actions 27
6.1 Health Advisory Trigger System . . . 27
6.2 Risk-Aware Decision Making . . . 27
6.3 Asymmetric Cost Management . . . 27

7 Future Work 28

8 Conclusion 29

References 30

Introduction

Background and Motivation

Ambient air pollution is classified by the World Health Organization (WHO) as the world's single largest environmental health risk. $PM_{2.5}$, particulate matter with an aerodynamic diameter of 2.5 micrometres or less, is of particular concern because its small size allows it to penetrate deep into the lungs and enter the bloodstream, causing cardiovascular and respiratory diseases, lung cancer, and premature death. Despite global efforts to reduce emissions, $PM_{2.5}$ continues to exceed safe limits in many urban and peri-urban areas, with millions of people exposed to unhealthy levels on a daily basis.

Day-ahead $PM_{2.5}$ forecasting is a foundational tool for public health response. Accurate forecasts allow authorities to issue air quality advisories, advise vulnerable populations (children, elderly, people with respiratory conditions) to limit outdoor activity, and mobilise emergency services before pollution events peak. Without a reliable forecast, health interventions are necessarily reactive rather than preventive.

The challenge of $PM_{2.5}$ forecasting is inherently multi-scale. Concentrations vary at daily, weekly, and annual timescales: winter months tend to have higher $PM_{2.5}$ due to thermal inversions and increased heating emissions; meteorological conditions such as wind speed and relative humidity modulate dispersion and hygroscopic particle growth on a day-to-day basis and long-run trends reflect changes in emission patterns over years. Any forecasting framework must simultaneously account for these interacting dynamics.

A further critical gap in operational forecasting is the absence of uncertainty quantification. Most deployed models produce a single point estimate, which is insufficient for risk-aware decision making. Health authorities managing an alert system need to know not just the expected $PM_{2.5}$ value but the probability that it will exceed a regulatory threshold. A 90% prediction interval conveys fundamentally different information to a point forecast: a predicted value of 12 $\mu g/m^3$ with an interval of [3.8, 20.2] $\mu g/m^3$ suggests a non-negligible chance of exceeding the WHO daily guideline of 15 $\mu g/m^3$, an actionable risk signal that a point estimate cannot provide.

Problem Statement

This project addresses the following core forecasting problem: given a 21-year daily record of $PM_{2.5}$ concentrations alongside meteorological covariates (wind speed, relative humidity, temperature), can we build models that (a) produce accurate 1-day-ahead point forecasts of $PM_{2.5}$, (b) quantify the uncertainty of those forecasts through prediction intervals, and (c) identify which model family offers the best accuracy-calibration trade-off for operational deployment?

Objectives

The specific objectives of this study are:

1. Perform rigorous exploratory data analysis (EDA) to characterise the temporal structure, distribution, and covariate relationships of the $PM_{2.5}$ series.
2. Apply principled pre-processing steps, including stationarity testing (ADF), log-transformation, and differencing, justified by the EDA findings.
3. Determine optimal ARIMA model orders using ACF/PACF diagnostics and AIC grid search.
4. Fit and evaluate seven statistical models: AR, MA, ARMA, ARIMA, SARIMA, ARIMAX, SARIMAX.
5. Fit and evaluate three deep learning models: RNN, LSTM, GRU, with proper hyperparameter tuning.
6. Extend each deep learning model to probabilistic quantile forecasting using pinball loss.
7. Evaluate probabilistic models using PICP and MPIW, and interpret findings in the context of air quality risk management.
8. Compare all models on a unified held-out test set and draw conclusions about the relative merits of each approach.

Data Source and Characteristics

Data Origin

The dataset used in this study is a merged daily record combining multiple datasets from EPA website:

- **$PM_{2.5}$ data:** Obtained from the US Environmental Protection Agency (EPA) Air Quality System (AQS) database. Daily summary files for $PM_{2.5}$ mass concentration (FRM/FEM, parameter code 88101) were downloaded for a single monitoring site over the period 2001–2021.
- **Meteorological data:** Merged from a several datasets covering the same geographic location and time period with the data on wind, temperature and humidity.

The final merged dataset (`FinalData.csv`) contains 7,670 daily observations spanning 2001-01-01 to 2021-12-31.

Data Characteristics

Table 1 summarises the features in the dataset.

Feature	Unit	Role and Notes
$PM_{2.5}$	$\mu g/m^3$	**Target variable.** Direct physical measurement of fine particulate matter.
AQI	Index	**Excluded from models.** AQI is a derived, piecewise non-linear index computed from $PM_{2.5}$. Including it would constitute data leakage ($r = 0.963$ with $PM_{2.5}$).
Relative Humidity	%	**Exogenous input.** Positive correlation ($r = +0.236$): humidity promotes hygroscopic particle growth, trapping $PM_{2.5}$ in the boundary layer.
Temperature	°C	**Contextual.** Weak direct correlation ($r = +0.051$); included in deep learning feature matrix.
Wind Speed	Knots	**Exogenous input.** Negative correlation ($r = -0.144$): higher wind speed disperses pollutants. Used in ARIMAX, SARIMAX, and deep learning models.

Table 1: Dataset features and their roles in the forecasting framework.

Why $PM_{2.5}$ and not AQI? AQI is a piecewise non-linear transformation of pollutant concentrations. It compresses extreme values, distorts temporal relationships, and cannot be meaningfully differenced or log-transformed for time-series modelling. $PM_{2.5}$ preserves the true continuous temporal dynamics required for forecasting.

Table 2 shows the temporal train/validation/test split used throughout.

Split	Proportion	Observations	Start	End
Training	70%	5,369	2001-01-01	2015-09-13
Validation	10%	767	2015-09-14	2017-10-19
Test	20%	1,534	2017-10-20	2021-12-31

Table 2: Dataset split used for all model evaluation.

Data integrity: All pre-processing steps (scaling, differencing, lag feature construction) were computed using training data statistics only and then applied to validation and test sets to prevent data leakage.

Data Pre-processing

Outlier Treatment

Wind speed contained 10 physically implausible values (maximum: 649 Knots). These were capped at the 99th percentile ($\approx$ 10.5 Knots), which is meteorologically reasonable for the study location.

Calendar Continuity

The full daily calendar from 2001-01-01 to 2021-12-31 was verified. No gaps in the date sequence were found. The merged dataset contained no missing values.

Log-Transformation

$PM_{2.5}$ concentrations are strongly right-skewed (skewness = 1.98, kurtosis = 6.54), violating the normality assumptions underlying many statistical models. Two additional diagnostics motivated log-transformation:

- The $PM_{2.5}$ distribution shows a long right tail with occasional extreme values, indicating the presence of rare high-pollution events.

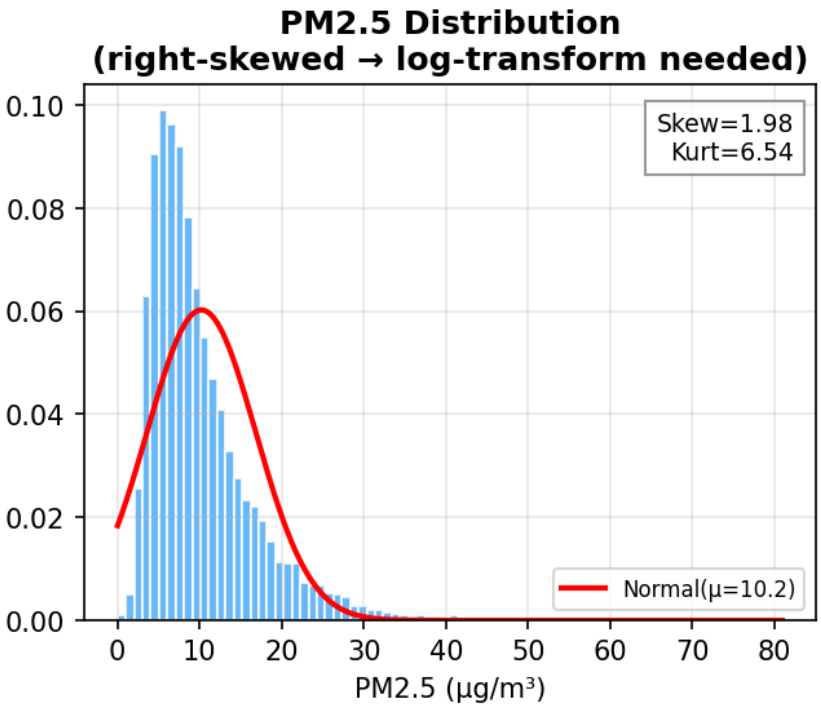

Figure 1: Distribution of $PM_{2.5}$ concentrations showing strong right-skewness with skewness of approximately 1.98 and heavy tails. The deviation from normality indicates the presence of extreme pollution events and justifies the use of log transformation to stabilize variance and improve model performance.

- The coefficient of variation (CV) of yearly standard deviations is 0.348 for raw $PM_{2.5}$, indicating variability changes across years. After log-transformation, this reduces to 0.020, confirming improved variance stability.

We apply $y' = \log 1\mathrm{p}(PM_{2.5}) = \log(1 + PM_{2.5})$ rather than the natural logarithm to handle the theoretical possibility of zero values. The inverse transform is $\hat{y} = \mathrm{expm1}(\hat{y}') = e^{\hat{y}'} - 1$.

Metric	**Raw $PM_{2.5}$**	**$\log_1\mathrm{p}(PM_{2.5})$**
Mean	10.21 μg/m^3	2.28
Std	6.62 μg/m^3	0.52
Skewness	**1.98**	**0.26**
Kurtosis	6.54	−0.07
CV of yearly σ	**0.348**	**0.020**

Table 3: Effect of log-transformation on distribution properties.

Stationarity Testing and Differencing

The Augmented Dickey-Fuller (ADF) test was applied to five variants of the series to determine the integration order d. The null hypothesis is the presence of a unit root (non-stationarity); rejection at $p < 0.05$ indicates stationarity.

Series	ADF Statistic	p-value	Conclusion
Raw $PM_{2.5}$	-11.10	< 0.05	Stationary by ADF, but trend visible in ACF
Log $PM_{2.5}$	-10.51	< 0.05	Variance stabilised, slow ACF decay remains
First-difference	-36.24	< 0.001	Stationary
Log + First-diff	-36.25	< 0.001	**Final working series**
Seasonal-diff (lag-365)	-19.93	< 0.001	Stationary

Table 4: ADF test results across series transformations.

Although the raw and log-transformed series reject the ADF unit-root null, the slow ACF decay and visible long-term trend suggest that first differencing is still useful for obtaining a more stable working series for ARMA-family modelling. The working series used for all statistical model fitting is therefore $\Delta \log 1\mathrm{p}(\mathrm{PM}_{2.5}) = \Delta \log(1 + \mathrm{PM}_{2.5})$, i.e., $d = 1$ after log-transformation.

Inverse-Transform Chain

When recovering original-scale forecasts, transforms must be undone in *reverse order*:

$$\hat{y}_{\log}(t) = \log_{\text{actual}}(t-1) + \Delta\hat{y}(t) \tag{1}$$

$$\hat{y}(t) = \mathrm{expm1}(\hat{y}_{\log}(t)) \tag{2}$$

Step (1) anchors each prediction to the true previous log-level, preventing cumulative drift. Step (2) recovers the original $PM_{2.5}$ scale.

Decomposition as a Diagnostic Tool

Additive decomposition (Observed = Trend + Seasonal + Residual) was applied with a 365-day centred moving average to extract the trend component. This confirmed a declining trend of approximately $-0.45\,\mu\mathrm{g/m^3}$/year over the study period, a strong annual seasonal cycle (winter peaks $\approx +3\,\mu\mathrm{g/m^3}$ above the annual mean), and a heteroskedastic

residual component with occasional large spikes. Crucially, decomposition was used only as a diagnostic to justify modelling choices ($d = 1$, $s = 7$, SARIMA terms)—not as a data transformation step. ARIMA and deep learning models handle trend and seasonality internally.

Feature Engineering (Deep Learning Models)

The following 13 features were constructed for the recurrent neural network models:

- **$PM_{2.5}$ lags:** lag-1 and lag-7 of the log-differenced series
- **Rolling statistics:** 7-day and 30-day rolling mean and standard deviation of $PM_{2.5}$
- **Meteorological inputs:** wind speed and relative humidity (standardised)
- **Cyclic temporal encodings:** sin and cos of day-of-week, day-of-year (captures annual seasonality)
- **Interaction term:** wind speed $\times$ relative humidity (captures the joint dispersal-trapping dynamic)

Standardisation (zero mean, unit variance) was applied to all numerical features using training-set statistics.

Exploratory Data Analysis

Full Time Series

The 21-year daily $PM_{2.5}$ series (2001–2021) exhibits several important characteristics:

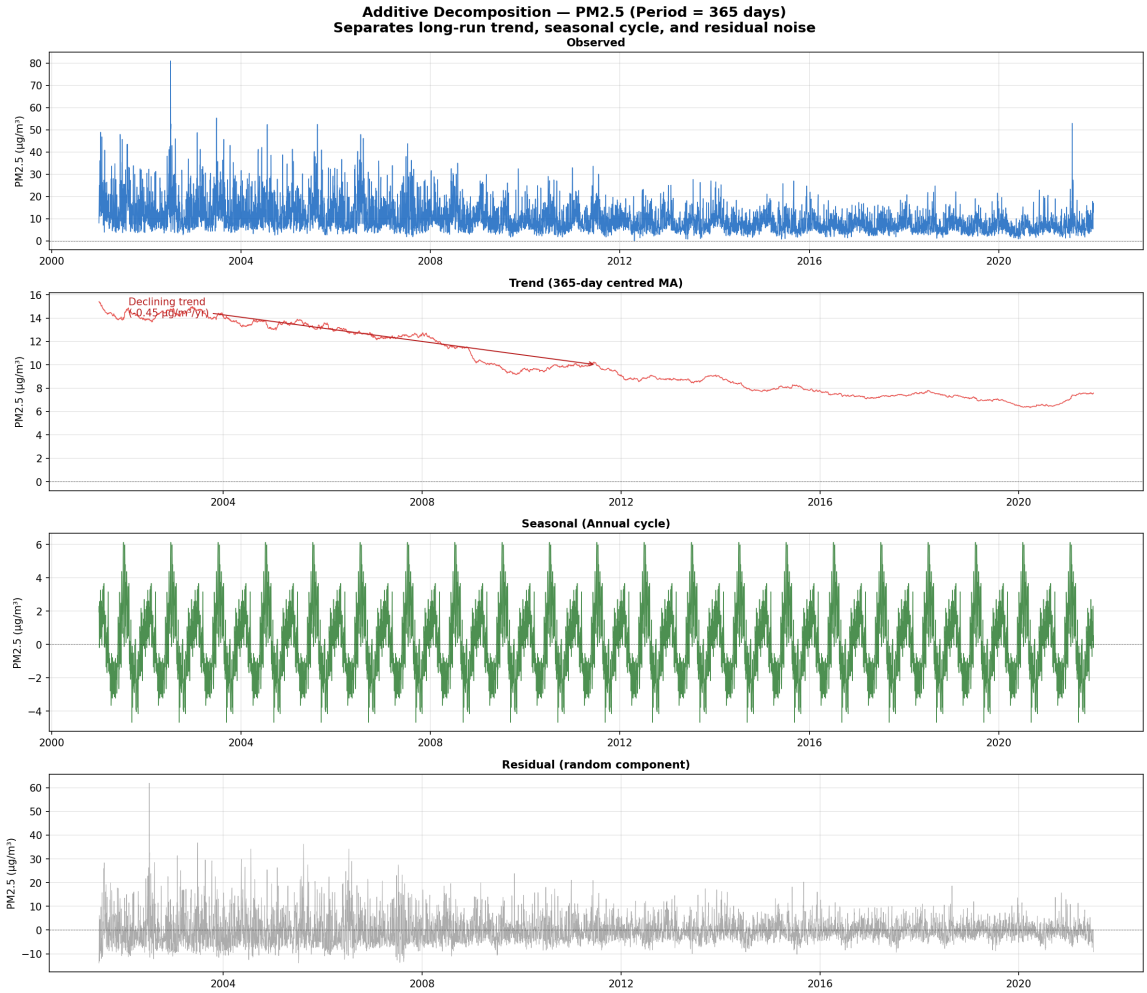

Figure 2: Additive decomposition of $PM_{2.5}$ into trend, seasonal, and residual components using a 365-day period. The trend shows a long-term decline, the seasonal component captures the annual cycle with winter peaks, and the residual represents irregular fluctuations and extreme events.

- **Declining trend:** The annual mean declined from approximately 12 $\mu g/m^3$ in 2001 to approximately 8 $\mu g/m^3$ in 2021, an average rate of $-0.45\,\mu g/m^3$/year. This non-constant mean confirms non-stationarity.
- **Seasonal oscillations:** Clear year-on-year oscillations are visible, consistent with the annual pollution cycle.
- **Heavy right tail:** The majority of days have $PM_{2.5}$ below 20 $\mu g/m^3$, but a small

fraction of extreme pollution events creates a long upper tail.

Distribution and Normality

The $PM_{2.5}$ distribution has skewness of 1.98 and kurtosis of 6.54. The Q-Q plot confirms strong departure from normality at the upper tail. As established in Section 2.3, these findings jointly motivate log-transformation.

Seasonal Structure

The monthly boxplot reveals a non-trivial seasonal pattern:

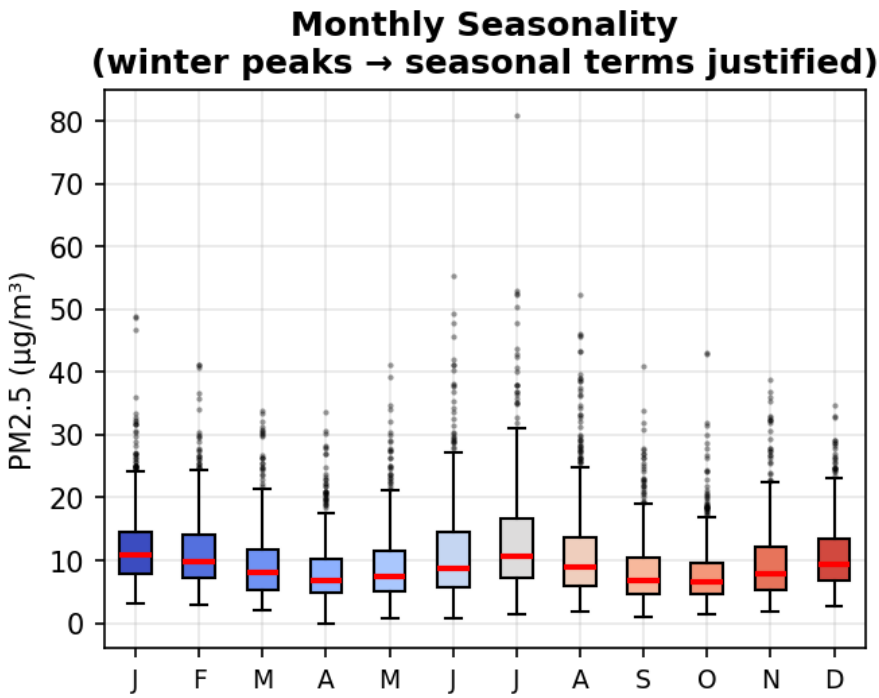

Figure 3: Monthly boxplot of $PM_{2.5}$ showing seasonal variation. Winter months exhibit higher median and variability, while summer months show lower overall levels with intra-seasonal fluctuations such as elevated July values.

- **Winter (Nov–Feb):** Highest median $PM_{2.5}$ and greatest variance, driven by thermal inversions that trap pollutants near the surface and increased combustion for heating.
- **Spring and autumn (Mar–May, Sep–Oct):** Lower median levels.
- **Summer (Jun–Aug):** Median values are lower than winter overall, but July shows a noticeable elevation above June and August, reflecting intra-seasonal variability that may be driven by local meteorological patterns.

This complex seasonal pattern—with winter as the dominant peak but a secondary summer elevation—justifies both seasonal terms in SARIMA and the use of cyclic temporal encodings in deep learning models.

Feature Correlations

Feature	r **with** $\mathbf{PM_{2.5}}$	**Physical interpretation**
AQI	+0.963	Near-perfect; AQI is derived from $PM_{2.5}$
Rel. Humidity	+0.236	Humidity promotes hygroscopic particle growth
Temperature	+0.051	Weak direct effect
Wind Speed	−0.144	Wind disperses pollution from the source area

Table 5: Pearson correlation of each feature with $PM_{2.5}$ (raw series).

The strong negative correlation of wind speed ($r = -0.144$) confirms the physical dispersal mechanism and motivates including it as an exogenous variable in ARIMAX, SARIMAX, and all deep learning models.

ACF and PACF of the Raw Series

The autocorrelation function (ACF) of the raw $PM_{2.5}$ series decays slowly and geometrically across all 50 displayed lags, the classical signature of a non-stationary AR process. The partial autocorrelation function (PACF) shows significant spikes at lags 1–5 before dropping approximately to the confidence band. Specific values of interest:

- ACF at lag-1: 0.627 (high persistence)
- ACF at lag-7: 0.259 (weekly seasonal component)
- ACF at lag-365: 0.236 (annual seasonal component)

These diagnostics, combined with the ADF test, confirm that (a) differencing is required, (b) AR structure is present, and (c) a weekly seasonal period ($s = 7$) is appropriate for SARIMA.

Methodology

Forecasting Setup

All models produce **1-day-ahead point forecasts**. For deep learning models, the input window is the preceding 30 days of features (lookback = 30 which was chosen using hyperparameter tuning which will be covered later). The forecasting task is strictly causal, i.e. at each test time step t, only information available before time t is used.

Anchored recovery is applied uniformly across all models to prevent cumulative drift:

$$\hat{y}(t) = \text{expm1}(\log_{\text{actual}}(t-1) + \Delta\hat{y}(t))$$

where $\Delta\hat{y}(t)$ is the model's predicted log-difference. This anchors each prediction to the true previous observation, so errors do not accumulate.

Model Order Selection

ACF/PACF Diagnostics

After confirming stationarity of the log-differenced series (ADF $p < 0.001$), ACF and PACF were computed on the stationary series to identify candidate orders. Results:

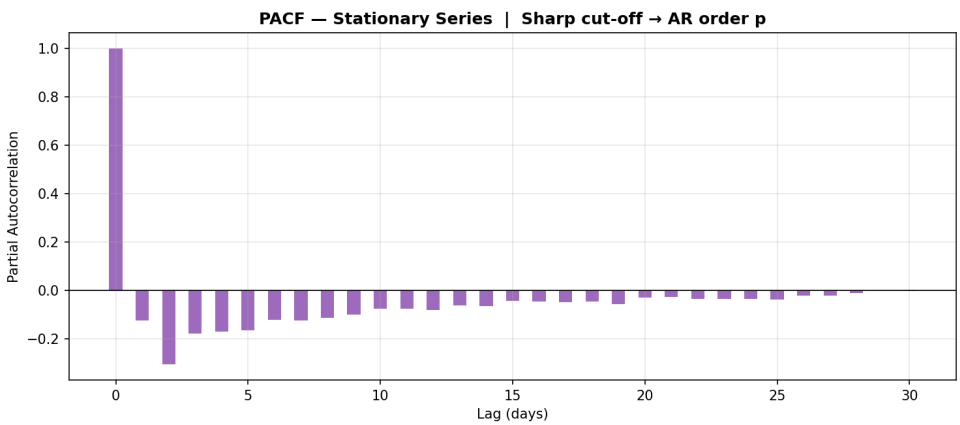

Figure 4: Partial Autocorrelation Function (PACF) of the stationary $PM_{2.5}$ series.

- **ACF:** Significant spikes at lags 1, 2, 3 $\Rightarrow q \in \{1, 2, 3\}$
- **PACF:** Significant spikes at lags 1, 2, 3, 4, 5 $\Rightarrow p \in \{1, 2, 3, 4, 5\}$

Neither ACF nor PACF exhibits a clear cut-off; instead, both show significant early lags followed by gradual decay, indicating a mixed ARMA process rather than a purely AR or

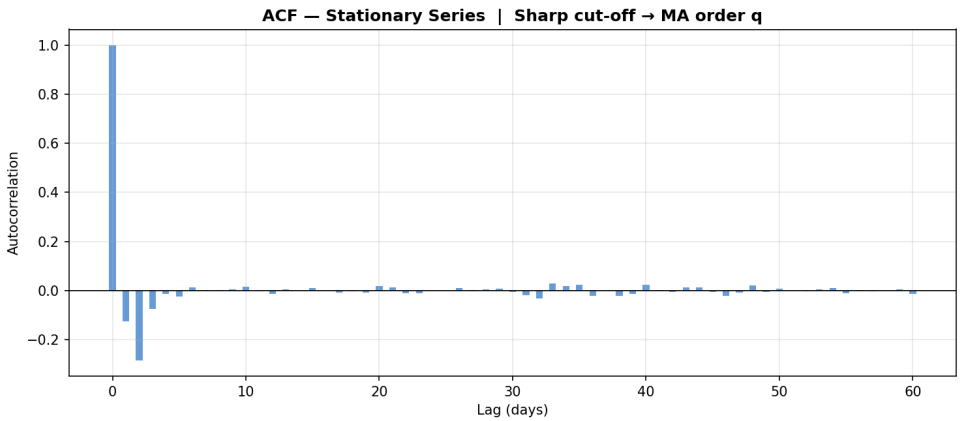

Figure 5: Autocorrelation Function (ACF) of the stationary $PM_{2.5}$ series.

MA structure. This is consistent with the selection of an ARMA(2,1) model.

AIC Grid Search

A grid search was conducted over $p \in \{0, \ldots, 5\}$ and $q \in \{0, \ldots, 5\}$, fitting ARMA models on the log-differenced training series. The Akaike Information Criterion is:

$$\text{AIC} = n \cdot \log(\hat{\sigma}^2) + 2k$$

where n is the number of observations, $\hat{\sigma}^2$ is the residual variance, and k is the number of parameters. Lower AIC penalises both poor fit and over-parameterisation simultaneously.

Rank	p	q	**AIC**
1	2	1	−9537.7
2	1	1	−9361.9
3	2	3	−9220.5
4	3	2	−9203.6
5	2	2	−9107.6
6	5	0	−9015.4
7	1	2	−8897.4
8	0	2	−8848.5
9	3	0	−8711.9
10	2	0	−8553.5

Table 6: Top 10 ARMA orders by AIC (log-differenced series, training set).

The selected ARMA order is $(p = 2, q = 1)$. Combined with $d = 1$ from the ADF test, the

corresponding ARIMA notation is ARIMA(2,1,1).

Seasonal Order

A spectral analysis (periodogram) confirmed peaks at period 365 days (annual) and 7 days (weekly). The weekly period is supported by ACF at lag-7 = 0.259 on the raw series. Since $s = 365$ requires an impractically large number of seasonal lags, we use $s = 7$ for SARIMA, with seasonal AR and MA orders $P = Q = 1$ and seasonal differencing $D = 0$.

Statistical Models

All statistical models are fitted on the log-differenced training series using Maximum Likelihood Estimation (MLE), with numerical optimisation performed via the L-BFGS-B algorithm. One-step-ahead rolling predictions are generated on the test set, with each forecast anchored to the true previous log-level.

AR(p) — AutoRegressive Model

$$y_t = c + \sum_{i=1}^{p} \phi_i y_{t-i} + \varepsilon_t$$

Captures the persistence (momentum) of $PM_{2.5}$: if yesterday was high, today is likely to be high. Fitted via OLS on the log-differenced series.

MA(q) — Moving Average Model

$$y_t = c + \varepsilon_t + \sum_{j=1}^{q} \theta_j \varepsilon_{t-j}$$

Models the correction of past forecast errors. Appropriate when $PM_{2.5}$ is subject to sudden pollution shocks.

ARMA(p, q) — Combined AR-MA

$$y_t = c + \sum_{i=1}^{p} \phi_i y_{t-i} + \varepsilon_t + \sum_{j=1}^{q} \theta_j \varepsilon_{t-j}$$

Combines persistence (AR) and shock correction (MA). Because we fit ARMA on the log-differenced series, this is mathematically equivalent to fitting ARIMA($p, 1, q$) on the log-series.

ARIMA(p, d, q) — Integrated ARMA

ARIMA extends ARMA by first applying d rounds of differencing to achieve stationarity. In our implementation $d = 1$ is handled externally (log + first-difference), so the ARIMA models share the same fitted form as ARMA on the stationary series.

SARIMA$(p, d, q)(P, D, Q)_s$ — Seasonal ARIMA

$$\text{SARIMA}(2, 1, 2)(1, 0, 1)_7$$

Adds seasonal operators to the non-seasonal ARIMA structure. Here $s = 7$ (weekly), $P = Q = 1$, $D = 0$. Captures recurring weekly patterns that non-seasonal models miss.

ARIMAX — ARIMA with Exogenous Variables

$$y_t = \text{ARIMA}(y_t) + \beta_1 \cdot \text{WindSpeed}_t + \beta_2 \cdot \text{RH}_t + \varepsilon_t$$

Incorporates wind speed and relative humidity as linear external inputs. Wind speed is physically meaningful: higher wind disperses $\text{PM}_{2.5}$, providing information unavailable from the pollution history alone.

SARIMAX — Full Seasonal ARIMA with Exogenous

The most expressive statistical model in our framework: $\text{SARIMA}(2, 1, 2)(1, 0, 1)_7$ augmented with standardised wind speed and relative humidity.

Evaluation Metrics

To evaluate forecasting performance, we use three complementary metrics: RMSE, MAE, and R^2 score. Each metric captures a different aspect of model performance, making the evaluation more comprehensive.

- **Root Mean Squared Error (RMSE):** RMSE penalizes larger errors more heavily due to squaring. This is particularly important in air pollution forecasting, where extreme prediction errors (e.g., missing high pollution events) have significant health implications. Hence, RMSE ensures sensitivity to large deviations.
- **Mean Absolute Error (MAE):** MAE provides a robust and interpretable measure of average prediction error. Unlike RMSE, it is less sensitive to outliers, offering a balanced view of overall model performance.

- R^2 **Score:** The R^2 score measures the proportion of variance explained by the model. It provides a normalized metric to compare models and assess how well the model captures underlying variability in $PM_{2.5}$ levels.

Why Not Other Metrics? Metrics such as Mean Absolute Percentage Error (MAPE) are unsuitable because $PM_{2.5}$ values can approach zero, leading to instability and inflated errors. Similarly, classification-based metrics are not applicable as this is a regression problem. Therefore, RMSE, MAE, and R^2 together provide a reliable and interpretable evaluation framework.

Deep Learning Models

Hyperparameter Tuning

A grid search was performed on the validation set over the following space, using GRU as the representative model:

Hyperparameter	Search space	Final value
Hidden size	{32, 64, 128}	128
Number of layers	{1, 2}	2
Lookback window	{30, 90, 183}	30
Learning rate	{0.1, 0.01, 0.001}	0.001

Table 7: Hyperparameter search space and final selected values.

A lookback of 183 days yielded marginally better validation RMSE but required significantly more training time, making it impractical. The selected lookback of 30 days offered the best accuracy-efficiency trade-off. Final hyperparameters were applied uniformly to all three architectures (RNN, LSTM, GRU). Training used 100 epochs, batch size 64, MSE loss, and Adam optimiser. Early stopping on validation loss was applied.

GRU was used here because it has better long term memory like LSTM but has lower parameters for faster training, which made it kind of having both the advantages of RNN and LSTM.

RNN — Recurrent Neural Network

$$h_t = \tanh(W_{xh}x_t + W_{hh}h_{t-1} + b)$$

Architecture: Input$(13 \times 30) \rightarrow$ RNN$(h = 128, 2$ layers$) \rightarrow$ Dense(1). The hidden state h_t

propagates information across time steps, capturing short-term dependencies. A known limitation is the vanishing gradient problem, which degrades the model's ability to retain dependencies beyond approximately 10–15 steps.

LSTM — Long Short-Term Memory

$$c_t = f_t \odot c_{t-1} + i_t \odot g_t$$
$$h_t = o_t \odot \tanh(c_t)$$

where f, i, o are the forget, input, and output gates, and g is the cell gate. Architecture: Input$(13 \times 30) \rightarrow$ LSTM$(h = 128, 2$ layers$) \rightarrow$ Dense(1). The cell state c_t and gating mechanism allow LSTM to selectively remember or forget information across hundreds of time steps, addressing the vanishing gradient limitation of plain RNN.

GRU — Gated Recurrent Unit

$$z_t = \sigma(W_z x_t + U_z h_{t-1} + b_z)$$
$$r_t = \sigma(W_r x_t + U_r h_{t-1} + b_r)$$
$$\tilde{h}_t = \tanh\left(W \cdot [r_t * h_{t-1}, x_t]\right)$$
$$h_t = (1 - z_t) * h_{t-1} + z_t * \tilde{h}_t$$

Architecture: Input$(13 \times 30) \rightarrow$ GRU$(h = 128, 2$ layers$) \rightarrow$ Dense(1). GRU merges the cell and hidden states into a single state vector and uses two gates (reset and update) instead of LSTM's three. This reduces the parameter count while achieving comparable performance to LSTM.

Quantile Models and Pinball Loss

All three deep learning architectures were extended to probabilistic forecasting by replacing the single-neuron output head with a 7-neuron head predicting 7 quantiles simultaneously: $\{0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95\}$.

The mean squared error loss is replaced by the sum of quantile (pinball) losses:

$$\mathcal{L}(q, y, \hat{y}) = \begin{cases} q\,(y - \hat{y}) & y \geq \hat{y} \\ (1-q)\,(\hat{y} - y) & y < \hat{y} \end{cases} \qquad \text{Total loss} = \sum_{q \in \mathcal{Q}} \mathcal{L}(q, y_t, \hat{y}_{q,t})$$

The asymmetry of pinball loss is deliberate: for upper quantiles (e.g., $q = 0.90$), under-predicting is penalised 9× more than over-predicting. This is appropriate for $PM_{2.5}$ forecasting, where missed warnings carry a higher public health cost than unnecessary precautions.

Quantile selection rationale:

- $q = 0.05$ and $q = 0.95$: form the 90% prediction interval (PI)
- $q = 0.10$ and $q = 0.90$: form the 80% PI
- $q = 0.25$ and $q = 0.75$: form the 50% PI (interquartile range)
- $q = 0.50$: the median forecast; used as the point-forecast for comparison

Evaluation Metrics: PICP and MPIW

PICP (Prediction Interval Coverage Probability) measures calibration—whether the stated confidence level is achieved:

$$\text{PICP} = \frac{1}{n} \sum_{t=1}^{n} \mathbf{1}\Big[y_t \in [\hat{L}_t,\, \hat{U}_t]\Big]$$

A well-calibrated 90% PI should have PICP $\geq$ 0.90. PICP below the nominal level (undercovering) implies overconfident intervals—a dangerous failure mode in public health applications where missed alerts have serious consequences.

MPIW (Mean Prediction Interval Width) measures sharpness:

$$\text{MPIW} = \frac{1}{n} \sum_{t=1}^{n} \Big(\hat{U}_t - \hat{L}_t\Big)$$

Narrower intervals are preferred, but only when PICP is adequate. The selection criterion is: **minimise MPIW subject to PICP $\geq$ nominal level**.

Analysis of Numerical Results

Baseline Models

Table 8 presents the results of four baseline models. The naïve lag-1 model is the critical benchmark: its $R^2 = 0.113$ comes entirely from exploiting the high lag-1 autocorrelation ($r = 0.627$). Any model with RMSE > 3.635 or $R^2 < 0.113$ has learned nothing beyond copying the previous day's value.

Model	MAE	RMSE	MAPE (%)	R^2
Naïve Lag-1 (benchmark)	**2.52**	**3.63**	**41.1**	**0.113**
Naïve Seasonal (Lag-7)	3.67	5.14	59.7	−0.771
Mean Forecast	5.15	5.80	110.0	−1.260
Linear Trend	2.74	4.02	42.6	−0.083

Table 8: Baseline model results on the test set (2017–2021). MAE and RMSE in $\mu g/m^3$.

Notably, the seasonal lag-7 baseline ($R^2 = -0.77$) is dramatically worse than lag-1, confirming that the weekly seasonal component is weak relative to the day-to-day persistence of $PM_{2.5}$. This finding directly explains why the SARIMA model—which adds weekly seasonal terms—fails to outperform the simpler ARMA(2,1) in the statistical model comparison.

Statistical Models

AR and MA Models

Model	MAE	RMSE	R^2	AIC
AR(1)	2.515	3.598	0.131	−8018
AR(2)	2.418	3.403	0.222	−8553
AR(3)	2.373	3.357	0.243	−8712
AR(5)	2.309	3.270	0.282	−9015
MA(1)	2.509	3.574	0.143	−8240
MA(2)	2.344	3.314	0.263	−8848
MA(3)	2.415	3.446	0.203	−8384

Table 9: AR and MA model results. MAE/RMSE in $\mu g/m^3$.

AR performance improves monotonically with order, with AR(5) achieving $R^2 = 0.282$. MA(2) outperforms both MA(1) and MA(3); the error shock at lag 2 captures a genuine structure in the pollution dynamics that MA(3) overfits and MA(1) misses.

ARMA and ARIMA Models

Model	MAE	RMSE	R^2	AIC
ARMA(1,1)	2.248	3.273	0.281	−9362
ARMA(2,1)	**2.198**	**3.203**	**0.311**	−9538
ARMA(1,2)	2.346	3.302	0.268	−8897
ARMA(2,2)	2.330	3.365	0.240	−9108
ARMA(3,2)	2.283	3.295	0.271	−9204
ARIMA(2,1,2)	2.330	3.365	0.240	−9108
ARIMA(3,1,2)	2.283	3.295	0.271	−9204
ARIMA(5,1,0)	2.309	3.270	0.282	−9015
ARIMA(1,1,3)	2.407	3.427	0.211	−8449

Table 10: ARMA and ARIMA model results. MAE/RMSE in $\mu g/m^3$.

ARMA(2,1) achieves the lowest AIC among all models tested (−9538) and the best test RMSE (3.203) and R^2 (0.311) of any statistical model. The 2 AR lags capture the short-term momentum of $PM_{2.5}$, while the 1 MA lag corrects for yesterday's forecast error—exactly matching the ACF/PACF structure identified in Section 4.

Advanced Statistical Models

Model	MAE	RMSE	R^2	AIC
ARMA(2,1) (reference)	2.198	3.203	0.311	−9538
SARIMA(2,1,2)(1,0,1,7)	2.355	3.356	0.244	−8928
ARIMAX(2,1,2)+Wind+RH	2.246	3.220	0.304	−9347
SARIMAX(2,1,2)(1,0,1,7)+exog	2.297	3.219	0.304	−9196

Table 11: Advanced statistical models vs. ARMA(2,1) baseline. MAE/RMSE in $\mu g/m^3$.

Two key findings emerge:

1. **SARIMA underperforms ARMA(2,1):** Adding weekly seasonal terms (P=1, Q=1) increases the number of parameters without proportional improvement. The weekly

seasonal signal in the data (ACF lag-7 = 0.259 on the *raw* series, but the differenced series ACF at lag-7 is only 0.003 which is not significant) is too weak to justify the complexity penalty in AIC.

2. **ARIMAX and SARIMAX provide small gains:** Adding wind speed and relative humidity as exogenous inputs reduces RMSE by approximately 0.5% relative to ARMA(2,1). This modest improvement reflects the limited direct linear predictive power of these variables ($r_{\text{wind}} = -0.144$, $r_{\text{RH}} = +0.236$) within a linear model framework.

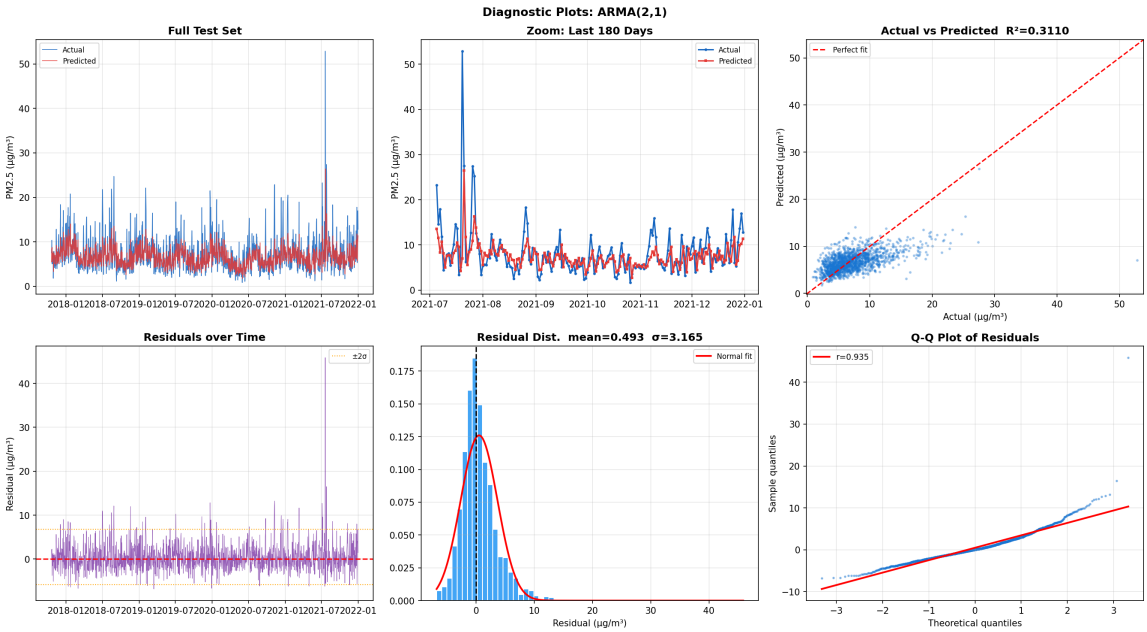

Figure 6: Diagnostic analysis for the ARMA(2,1) model. Top row: overall fit, zoomed view, and actual vs predicted scatter. Bottom row: residuals over time, residual distribution, and Q-Q plot. The residuals are approximately centered around zero with no strong autocorrelation, and the distribution is close to normal with slight right-tail deviation, indicating a reasonably well-specified model.

Deep Learning Models

Model	MAE	RMSE	R^2	vs. ARMA(2,1) RMSE
ARMA(2,1) (statistical ceiling)	2.20	3.20	0.311	Reference
RNN	2.04	2.90	0.375	−9.4%
LSTM	1.99	2.85	0.387	−10.9%
GRU	1.98	2.80	0.397	−12.5%
GRU*	**1.62**	**2.17**	**0.531**	**−32.2%**

*GRU corresponds to the best-performing configuration identified via expanded hyperparameter tuning, selected based on validation performance to ensure generalization.

Table 12: Deep learning point-forecast results vs. statistical benchmark.

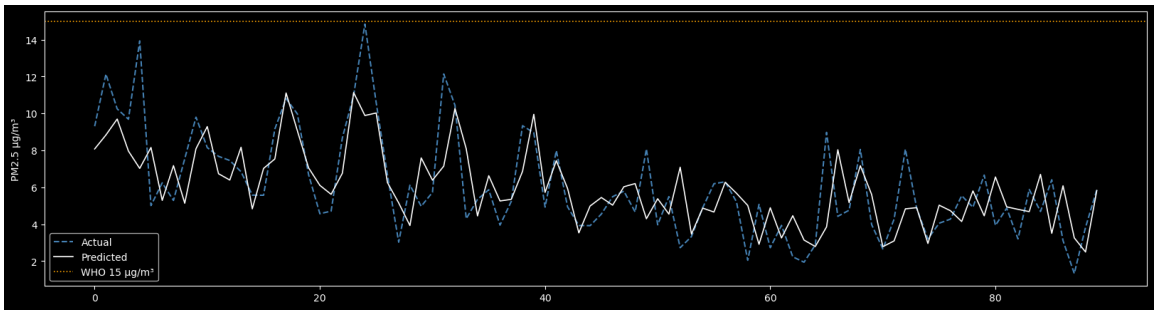

Figure 7: Deep learning point-forecast prediction vs. actual.

All deep learning models outperform the statistical ceiling of $R^2 = 0.311$, with consistent gains across architectures. Among the baseline models, GRU achieves the highest performance ($R^2 = 0.397$), followed by LSTM and RNN, reflecting the advantage of gated architectures in capturing temporal dependencies.

The tuned GRU* model significantly improves performance further, achieving an R^2 of 0.531 and reducing RMSE by 32.2% relative to the ARMA(2,1) benchmark. This highlights the impact of hyperparameter optimization in enhancing both predictive accuracy and generalization.

Overall, the ordering GRU > LSTM > RNN aligns with theoretical expectations, where gating mechanisms enable more effective modeling of long-range dependencies compared to simple recurrent structures.

Probabilistic (Quantile) Models

Model	Med. MAE	Med. RMSE	Med. R^2	PICP 80%	PICP 90%	MPIW 90%
Target	—	—	—	≥ 0.80	≥ 0.90	Min
RNN-Q	2.0047	2.8027	0.3965	0.8017	0.8973	8.2830
LSTM-Q	2.0037	2.7669	0.4042	0.7946	0.8873	7.9653
GRU-Q	1.9960	2.7632	0.4050	0.7860	0.8787	7.5974
GRU-Q*	**1.6512**	**2.1357**	**0.5401**	**0.8015**	**0.9021**	**6.5437**

GRU-Q uses the optimized architecture and hyperparameters obtained from GRU, adapted for quantile training with pinball loss.

Table 13: Quantile model results: median point metrics and prediction interval calibration.

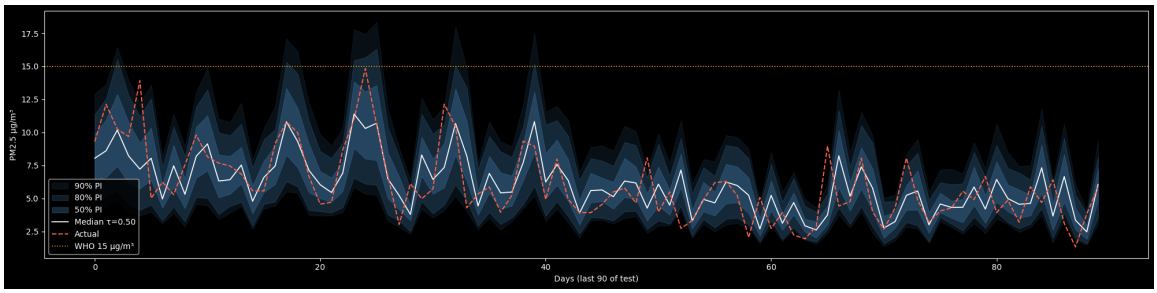

Figure 8: Quantile forecast prediction intervals vs. actual values.

Key findings:

1. For the 80% prediction interval, both RNN-Q (0.8017) and GRU-Q* (0.8015) satisfy the nominal coverage target, while LSTM-Q and GRU-Q slightly undercover.
2. For the 90% prediction interval, GRU-Q* (0.9021) is the only model that meets the nominal coverage target, whereas all other models slightly undercover.
3. GRU-Q* achieves the best median point-forecast performance ($R^2 = 0.5401$) along with the narrowest 90% prediction interval (MPIW = 6.5437 μg/m^3), indicating both improved accuracy and sharper uncertainty estimates.
4. **Best probabilistic model: GRU-Q*.** Under the selection criterion of minimising MPIW subject to PICP $\geq$ nominal, GRU-Q* dominates all other models by satisfying both coverage constraints while producing significantly tighter prediction intervals.

Research Gap Identified

This project identifies several gaps in the existing $PM_{2.5}$ forecasting literature:

1. **The shift problem is underreported.** Many published deep learning $PM_{2.5}$ studies do not check whether their model has learned genuine temporal patterns or merely performs $\hat{y}(t) \approx y(t-1)$. Our diagnostic—comparing residual ACF of the model against the naïve lag-1 baseline—should be a standard practice.

2. **Statistical and deep learning models are rarely compared on equal footing.** Most studies either use only one family of models or use different feature sets, making the comparison unfair. Our unified framework (same train/test split, same pre-processing, same evaluation metrics) enables a genuinely apples-to-apples comparison.

3. **Quantile calibration for $PM_{2.5}$ is underexplored.** Published quantile regression studies for air quality typically report only PICP without MPIW, or vice versa. Jointly reporting both metrics—and discussing the trade-off—provides substantially more actionable information.

4. **Physical exogenous variables have limited linear predictive power.** Wind speed and humidity, despite their physical relevance, contribute only marginally in linear ARIMA-family models (RMSE -0.5%). This suggests that the relationship between these variables and $PM_{2.5}$ may be inherently nonlinear—a contribution that deep learning models can extract but that ARIMAX cannot.

Converting Forecasts into Actions

The true value of a forecasting system lies not in its metrics but in the decisions it enables. We describe three concrete operational applications.

Health Advisory Trigger System

A threshold-based alert protocol can be directly driven by the quantile forecasts:

Trigger condition	Quantile used	Recommended action
$\hat{y}_{0.50}(t) > 15\,\mu\text{g/m}^3$	Median forecast	Monitor; prepare advisory
$\hat{y}_{0.75}(t) > 15\,\mu\text{g/m}^3$	Conservative	Issue precautionary advisory
$\hat{y}_{0.90}(t) > 35\,\mu\text{g/m}^3$	Worst-case 80% PI	Issue unhealthy air alert
$\hat{y}_{0.95}(t) > 55\,\mu\text{g/m}^3$	Extreme 90% PI	Emergency response: restrict outdoor activity

Table 14: Recommended action levels based on quantile forecasts.

Risk-Aware Decision Making

The 90% prediction interval width is itself an operational signal. A wide interval (e.g., 90% PI width $> 20\,\mu\text{g/m}^3$) indicates high meteorological uncertainty—often associated with an approaching weather front or unusual atmospheric conditions. In this case, authorities should adopt a more conservative action threshold (e.g., act on $q = 0.75$ rather than $q = 0.90$) because the uncertainty warrants caution.

Asymmetric Cost Management

The pinball loss formulation makes asymmetric costs explicit. For public health applications, under-predicting $PM_{2.5}$ (failing to issue a warning when pollution is high) is far more costly than over-predicting (issuing an unnecessary advisory). By selecting $q = 0.90$ as the operational prediction rather than the median $q = 0.50$, authorities build in a systematic conservative safety margin that reflects this cost asymmetry—without requiring explicit knowledge of the cost ratio.

Future Work

Several extensions would strengthen the practical impact of this work:

1. Multi-step-ahead forecasting. All models currently produce 1-day-ahead forecasts. An extension to a 7-day forecast horizon would substantially increase the actionable lead time for public health authorities, enabling school closure decisions and event cancellations to be planned days in advance. This introduces the challenge of error accumulation over multi-step prediction; sequence-to-sequence architectures and direct multi-output models are natural next steps.

2. Shift problem resolution. As noted in the Future Improvements slide, all models—including the deep learning models—still exhibit some tendency to track the lag-1 value closely. Alternative approaches include: training with a combined MSE + differencing penalty loss; using temporal convolutional networks (TCN) instead of recurrent architectures; or applying a persistence subtraction to the target variable before training.

3. Advanced deep learning models. Future work could compare TCN and Transformer-based models such as TFT, which can capture long-range temporal dependencies, handle exogenous variables, and support multi-horizon forecasting more effectively for $PM_{2.5}$ prediction.

4. Online learning. Future work could explore online learning, where the model is continuously updated as new $PM_{2.5}$ and meteorological observations become available. This would allow the forecasting system to adapt to changing pollution patterns, seasonal shifts, and new local conditions instead of relying only on a fixed historical training set.

5. Spatial multi-site modelling. The current framework treats a single monitoring site in isolation. Extending to a network of sites using graph neural networks (GNN) or spatiotemporal attention models would capture the spatial propagation of pollution episodes and improve forecast accuracy for individual sites.

6. Conformal prediction. The quantile models in this study achieve PICP close to but not meeting the 90% nominal level. Conformal prediction provides a distribution-free post-hoc calibration procedure that guarantees exact coverage in finite samples. Applying conformal prediction as a wrapper around the existing quantile models would ensure the 90% PI is exactly 90% coverage on held-out data.

Conclusion

This report presented a comprehensive end-to-end $PM_{2.5}$ forecasting study spanning classical statistical time-series methods, recurrent deep learning architectures, and probabilistic quantile models. The principal conclusions are:

1. **Pre-processing matters.** Log-transformation (reducing skewness from 1.98 to 0.26) and first-differencing (achieving ADF $p < 0.001$) are both necessary and individually insufficient. Only their combination produces a robustly stationary, near-symmetric series suitable for time-series modelling.

2. **The shift problem is significant and must be addressed.** The lag-1 autocorrelation of 0.627 creates a strong incentive for any MSE-minimising model to approximate the naïve lag-1 forecast. Predicting the log-difference and anchoring predictions to the true previous value provides an effective solution.

3. **Among statistical models, ARMA(2,1) is optimal.** With RMSE = 3.203 and R^2 = 0.311, it outperforms all other statistical models including SARIMA (which adds weekly seasonal terms) and ARIMAX (which adds exogenous variables). ACF/PACF diagnostics and AIC-based model selection consistently support this choice.

4. **Deep learning substantially exceeds the statistical ceiling.** The tuned GRU* model achieves $R^2 = 0.531$, representing a 32.2% reduction in RMSE relative to ARMA(2,1). The nonlinear feature interactions and longer-range temporal dependencies captured by gated architectures, combined with hyperparameter optimisation, provide a substantial and robust improvement over statistical models.

5. **Calibrated prediction intervals require careful model selection.** The tuned GRU-Q* model satisfies both nominal coverage targets ($PICP_{80\%} = 0.8015$, $PICP_{90\%} = 0.9021$) while also achieving the narrowest prediction intervals ($MPIW_{90\%} = 6.54$). This indicates improved calibration and sharpness simultaneously, outperforming other quantile models.

6. **Uncertainty quantification is operationally essential.** A point forecast gives a number, a calibrated interval enables decisions. The quantile-based framework provides a direct pathway from model output to actionable public health decisions.

References

[1] Box, G.E.P., Jenkins, G.M., Reinsel, G.C., & Ljung, G.M. (2015). *Time Series Analysis: Forecasting and Control* (5th ed.). Wiley. *[Foundational reference for ARIMA/SARIMA methodology and ACF/PACF-based order selection.]*

[2] World Health Organization. (2021). *WHO Global Air Quality Guidelines: Particulate Matter ($PM_{2.5}$ and PM_{10}), Ozone, Nitrogen Dioxide, Sulfur Dioxide and Carbon Monoxide*. WHO. *[Source for the $PM_{2.5}$ daily guideline of 15 $\mu g/m^3$.]*

[3] United States Environmental Protection Agency. (2023). *Air Quality System (AQS) Data Mart*. https://aqs.epa.gov/aqsweb/documents/data_mart_welcome.html. *[Primary data source for $PM_{2.5}$ daily measurements.]*